

Satellite Structural Inspection Interferometric Synthetic Aperture Radar (InSAR) Based

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METHODOLOGY

1. Satellite SAR System with Resolution and Revisit Time

To select a suitable SAR system, technical specifications such as spatial resolution, frequency band, polarization, and revisit time were considered. Sentinel-1, operating at C-band (5.405 GHz), was selected due to:

Parameter	Sentinel-1 C-SAR (IW Mode)
Frequency Band	C-band (~5.405 GHz)
Revisit Time	12 days (single satellite)
Spatial Resolution	5 m (range) × 20 m (azimuth)
Coverage Mode	Interferometric Wide Swath (IW)

2. Retrieving Multi-Temporal SAR Imagery Over the Specified Period

Multi-temporal **Sentinel-1 SLC IW mode** data were downloaded based on:

- **Area of Interest (AOI):** A polygon covering the bridge and surrounding terrain.
- **Date range:** Multiple acquisitions were collected spanning several months to capture temporal variations.
- **Image acquisition conditions:** All images were selected with the same relative orbit, pass direction (ascending or descending), and polarization (typically VV/VH) to ensure consistency for interferometric analysis.

S1A_IW_SLC__1SDV_20250111T130240_20250111T130310_057397_0710AB_813D
S1A_IW_SLC__1SDV_20250123T130239_20250123T130309_057572_07179E_5431
S1A_IW_SLC__1SDV_20250204T130238_20250204T130308_057747_071E86_3FE6
S1A_IW_SLC__1SDV_20250216T130238_20250216T130308_057922_072592_C312
S1A_IW_SLC__1SDV_20250228T130238_20250228T130307_058097_072CB4_7A14
S1A_IW_SLC__1SDV_20250312T130238_20250312T130308_058272_0733CA_217F
S1A_IW_SLC__1SDV_20250324T130238_20250324T130308_058447_073AB5_B445
S1A_IW_SLC__1SDV_20250405T130238_20250405T130308_058622_0741CE_2B52
S1A_IW_SLC__1SDV_20250417T130238_20250417T130308_058797_0748F4_3FFD
S1A_IW_SLC__1SDV_20250429T130239_20250429T130309_058972_07501E_08D8
S1A_IW_SLC__1SDV_20250511T130238_20250511T130308_059147_0756BA_DE13
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S1A_IW_SLC__1SDV_20250628T130236_20250628T130306_059847_076EE3_6BFD

3. Differential SAR Interferometry (D-InSAR) for Displacement Measurement

D-InSAR analysis was performed to measure millimeter-scale deformation over time:



VISUALIZATIONS

1. Defining specific bridge location & parameters for satellite SAR based inspection



Project & Bridge Details

State – Rajasthan

Bridge No. – 903/2

Bridge Chainage – 899+800

Project - Chittorgarh to Kota

Zone – RJ 06

Structure Type – MJB

Sar Parameters

Sensor – Sentinel-1 (C-Band)

Beam Mode – IW (Interferometric Wide Swath)

Path – 34

Frame – 519

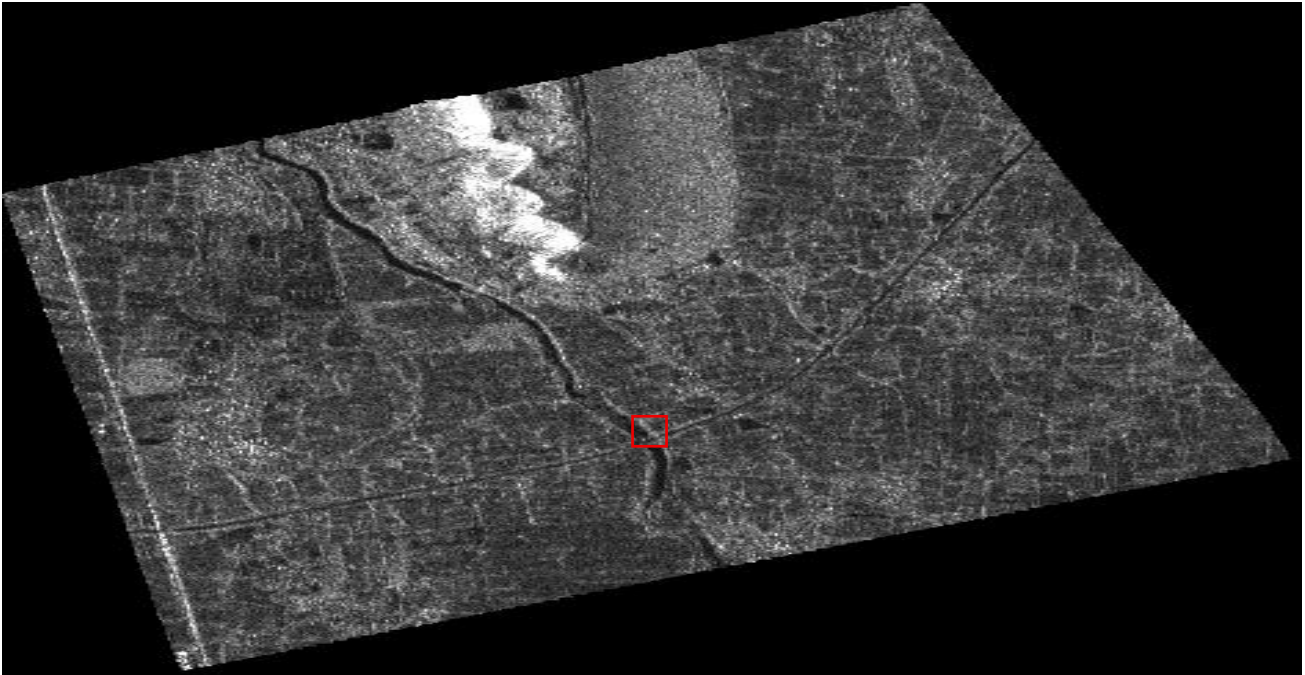
Flight Direction - DESCENDING

Polarization – VV+VH

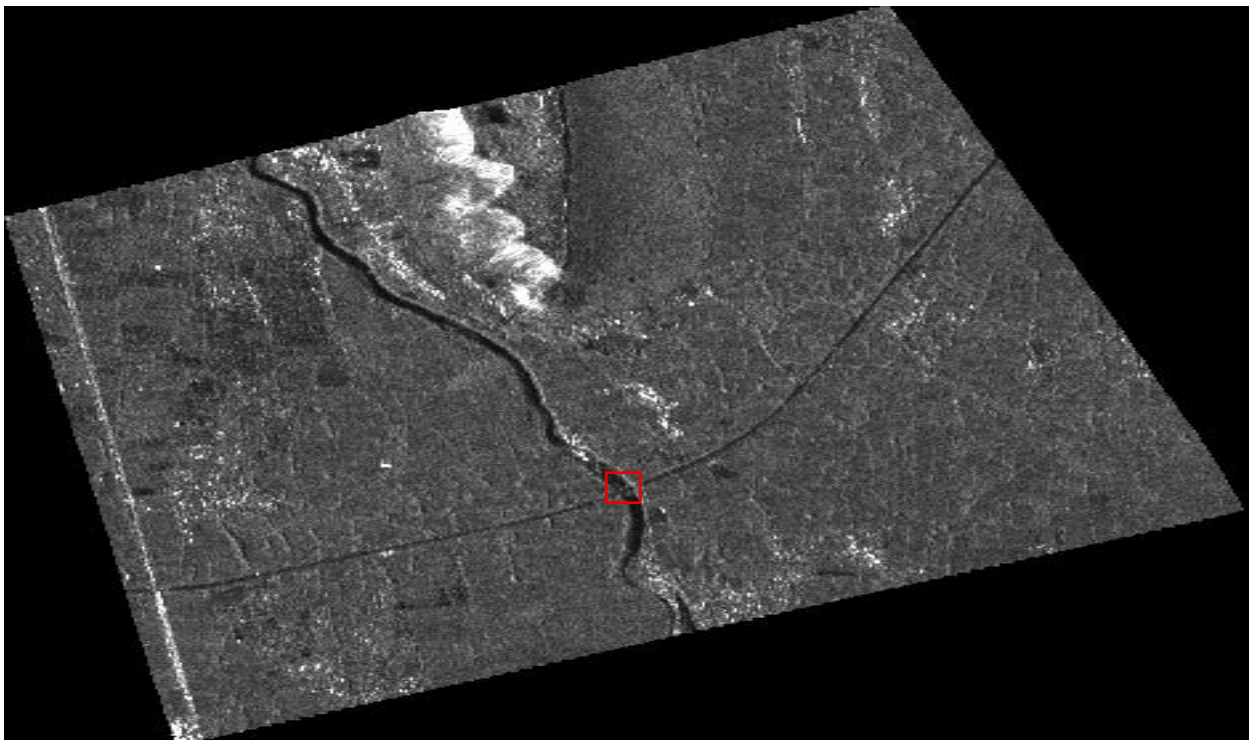
Absolute Orbit – 54706

2. Calibration, georeferencing, orthorectification of SAR images (VH/VV)

- Orthorectification of SAR image VH Polarization



- Orthorectification of SAR image VV Polarization

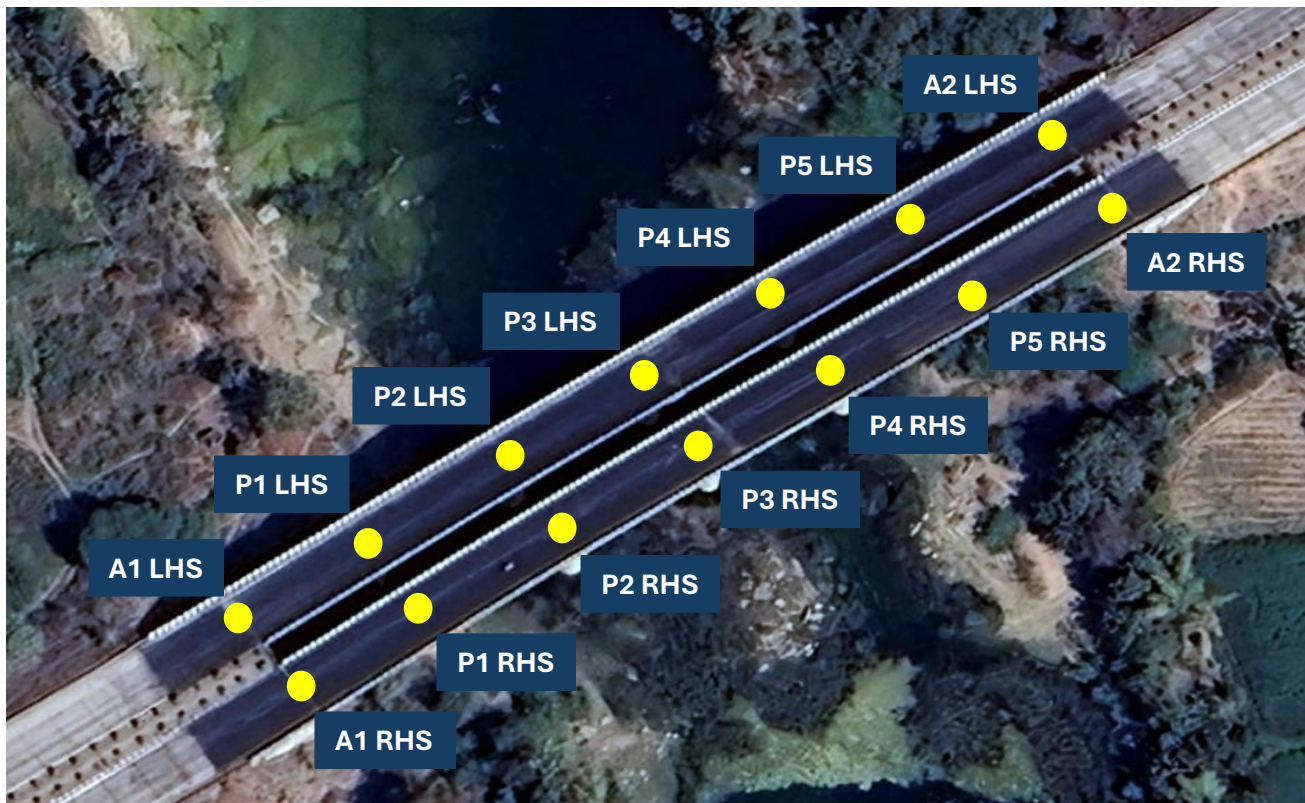


3. Results of SAR based bridge inspection, including visualizations

Bridge Components

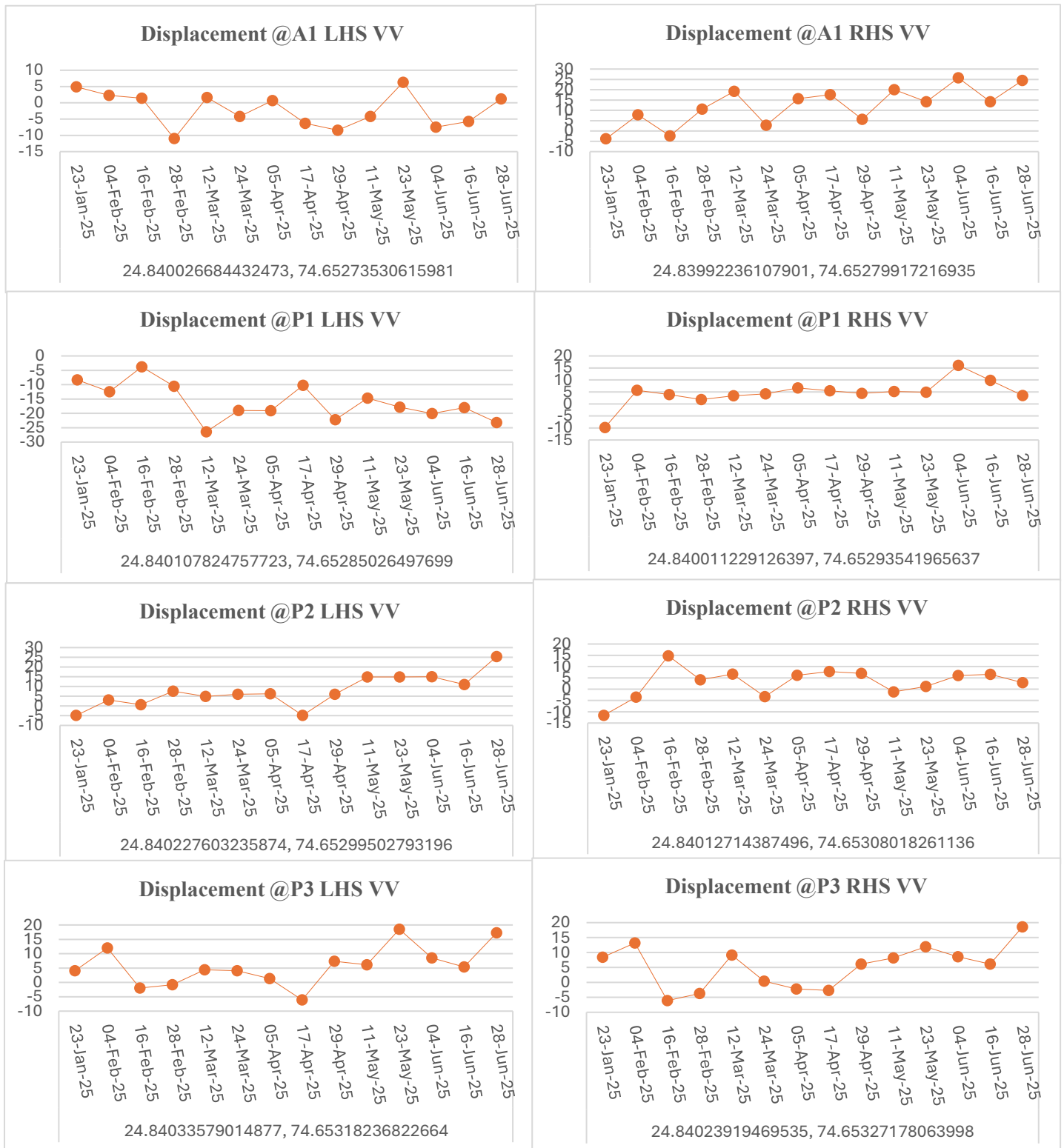
MJB – CH-899+800

Length – 135.20M

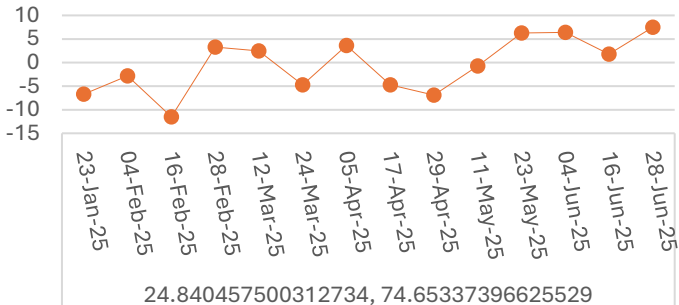


Scan Duration		From : 23 Jan 2025	To : 28 June 2025
Components & Abbreviation			
A1		Abutment 1	
P1		Pier 1	
P2		Pier 2	
P3		Pier 3	
P4		Pier 4	
P5		Pier 5	
A2		Abutment 2	
LHS		Left Hand Side	
RHS		Right Hand Side	

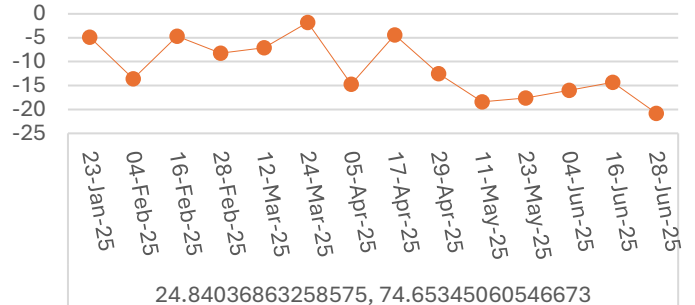
3.1 VV Displacement from 23 Jan 2025 to 28 June 2025



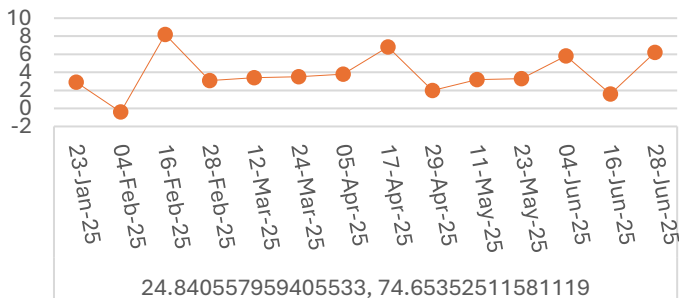
Displacement @P4 LHS VV



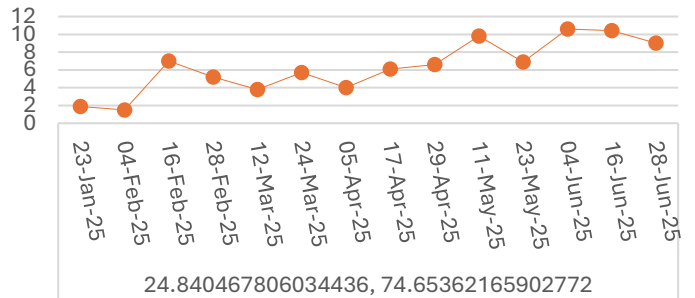
Displacement @P4 RHS VV/VH



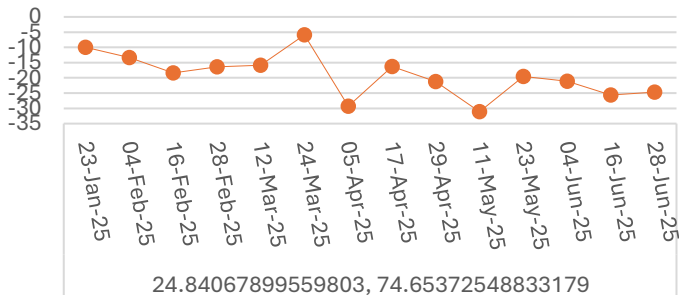
Displacement @P5 LHS VV



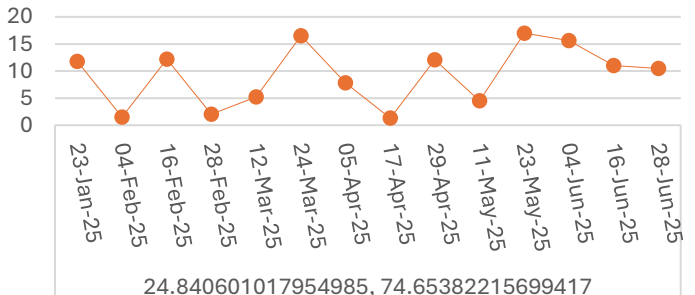
Displacement @P5 RHS VV



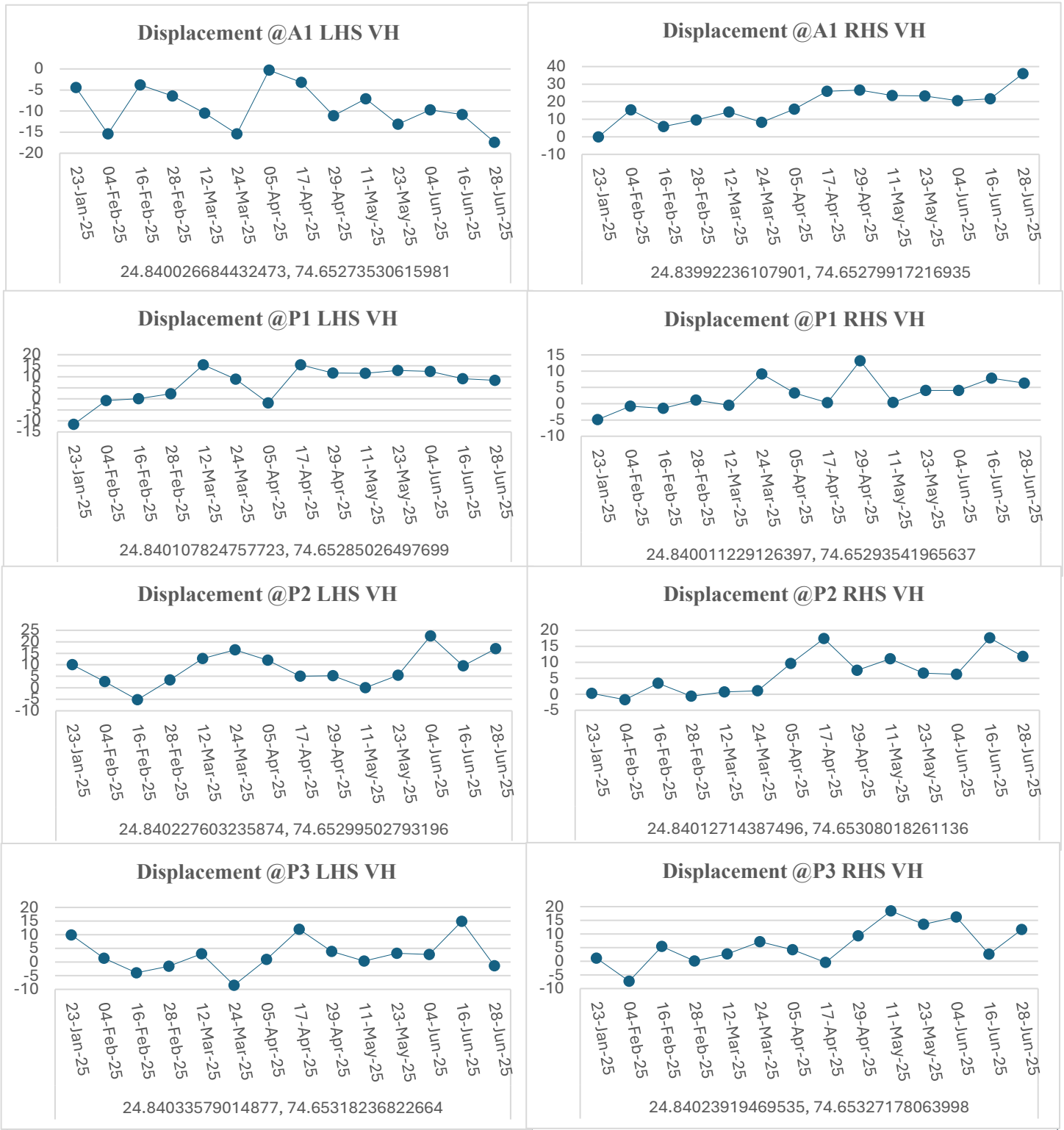
Displacement @A2 LHS VV



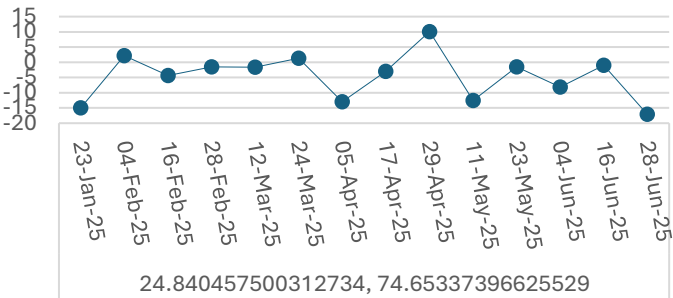
Displacement @A2 RHS VV



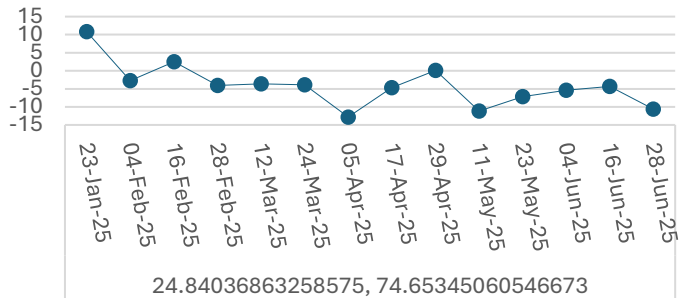
3.2 VH Displacement From 21 June 2024 to 24 March 2025



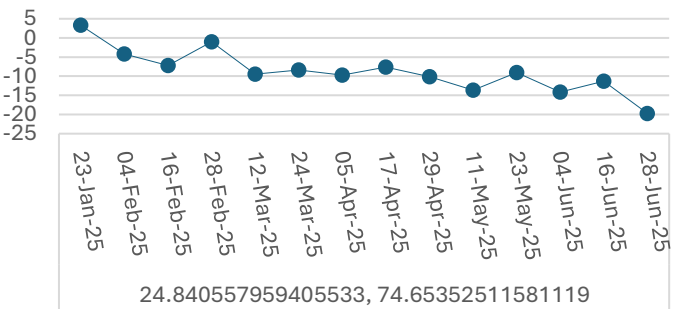
Displacement @P4 LHS VH



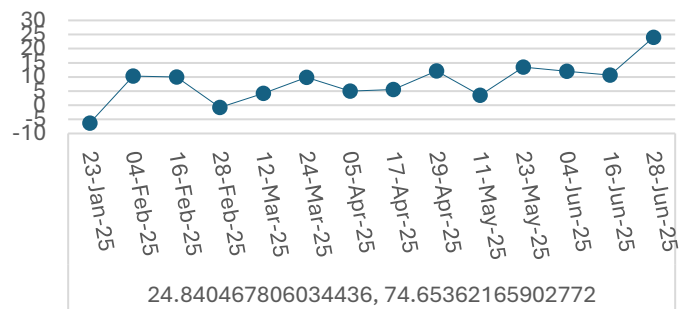
Displacement @P4 RHS VH



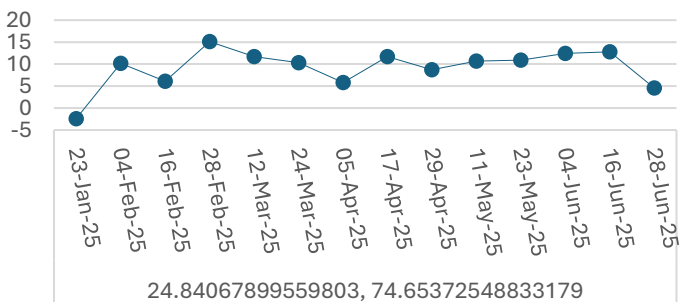
Displacement @P5 LHS VH



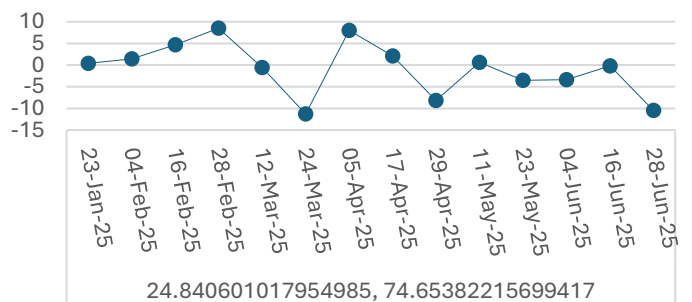
Displacement @P5 RHS VH



Displacement @A2 LHS VH



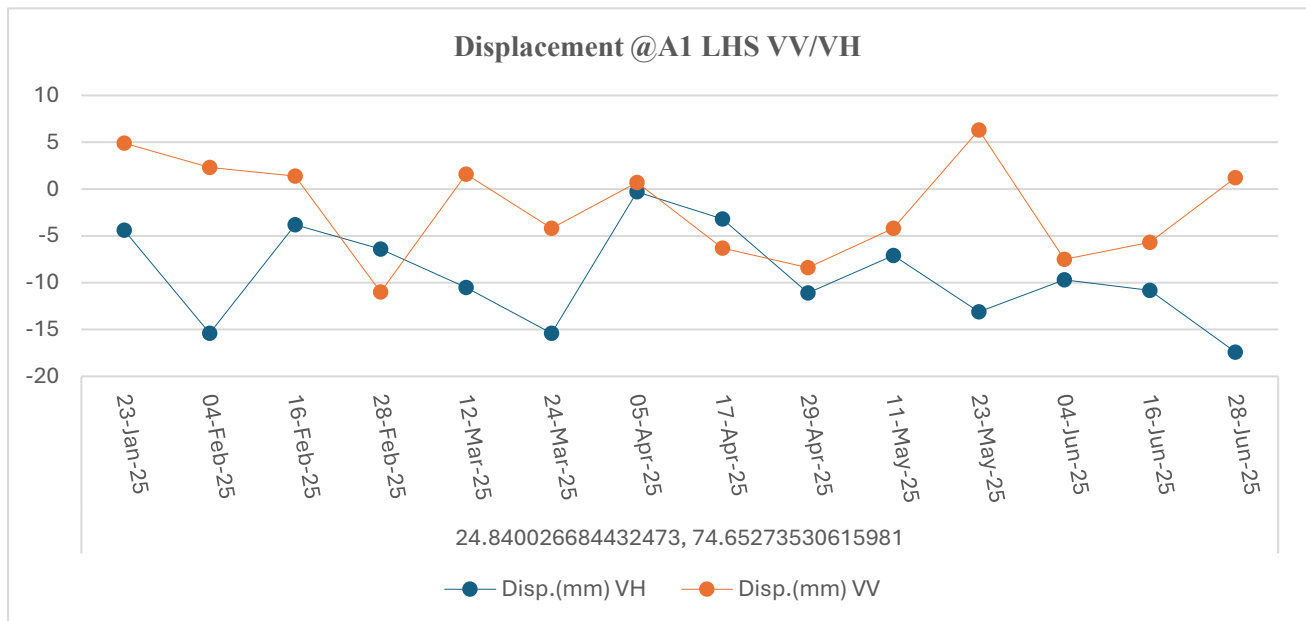
Displacement @A2 RHS VH



3.3 SAR Based Analysis at Abutment 1 LHS:



A1 LHS (24.840026684432473, 74.65273530615981)



Key Observations:

Displacement Magnitude Patterns:

- **Vertical Displacement (VV)** Vertical displacement shows mixed behaviour, fluctuating between approximately -10 mm and +6 mm.
- **Horizontal displacement (VH)** Horizontal displacement remains consistently negative throughout the monitoring period, indicating sustained movement in a single lateral direction.

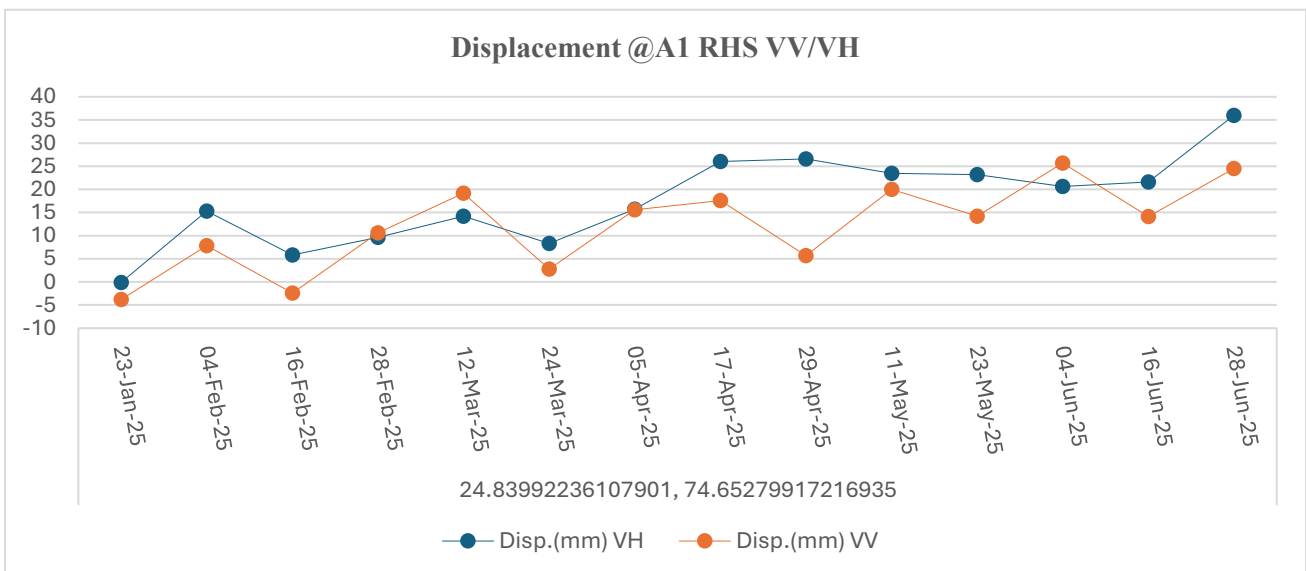
Critical Events & Anomalies:

- **February 2025:** VH drops sharply to approximately -15 mm, coinciding with a brief reduction in VV — indicating a short-term combined lateral and vertical response.
- **May 2025:** VV peaks near +6 mm, contrasting with a strong negative VH (-13 mm), highlighting opposing displacement behaviour between vertical and horizontal components.
- **June 2025:** VH reach one of its lowest value while VV stabilize indicating continue lateral drift dominance.

3.4 SAR Based Analysis at Abutment 1 RHS:



A1 RHS (24.83992236107901, 74.65279917216935)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal displacement (VH)** Horizontal displacement shows a predominantly positive trend, increasing from approximately +2 mm to nearly +35 mm over the monitoring period.
- **Vertical displacement (VV)** Vertical displacement exhibits moderate variability, ranging approximately between -3 mm and +25 mm. Early monitoring stages show small oscillations around zero, followed by progressive uplift from mid-March onwards.

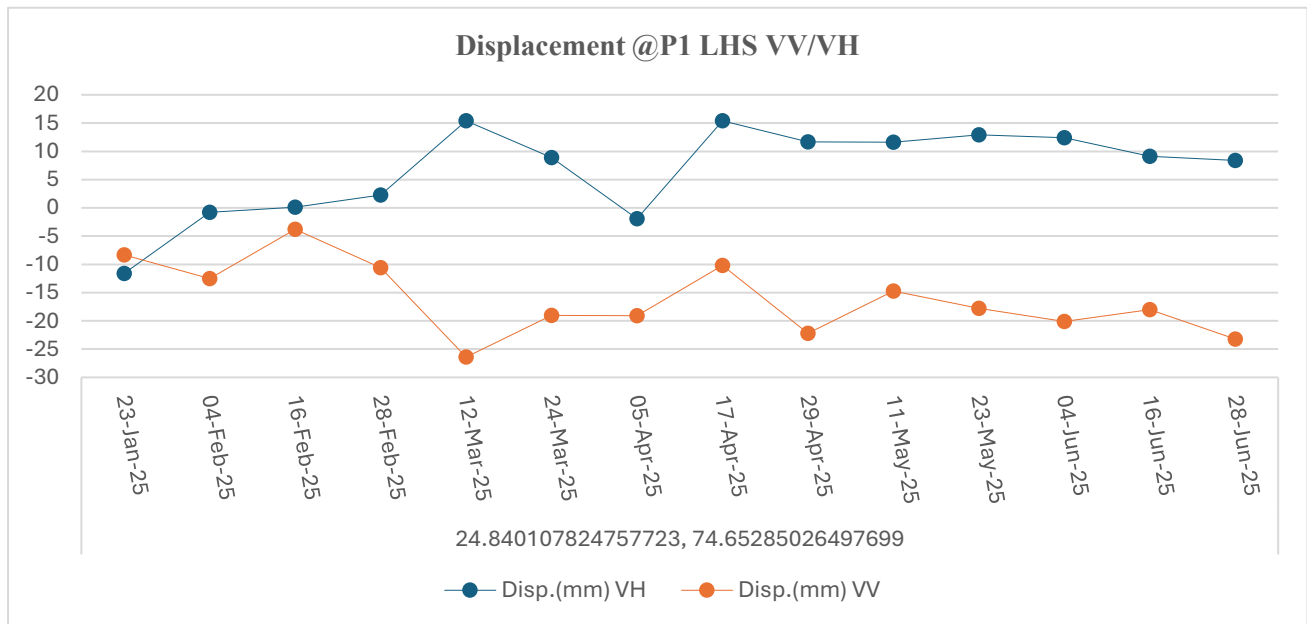
Critical Events & Anomalies:

- **Feb 2025:** Both VH and VV display minor fluctuations, indicating relatively stable initial conditions.
- **May 2025:** VH remains elevated while VV temporarily drops, indicating temporary decoupling between lateral vertical responses
- **June 2025:** VH reaches its maximum recorded value accompanied by sustained positive VV indicating significant cumulative deformation

3.5 SAR Based Analysis at Pier 1 LHS:



P1 LHS (24.840107824757723, 74.65285026497699)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal Displacement (VH)** Horizontal displacement shows a predominantly positive trend, ranging approximately from -10 mm to +15 mm.
- **Vertical Displacement (VV)** Vertical displacement is consistently negative, ranging approximately from -8 mm to -25 mm, indicating sustained downward movement.

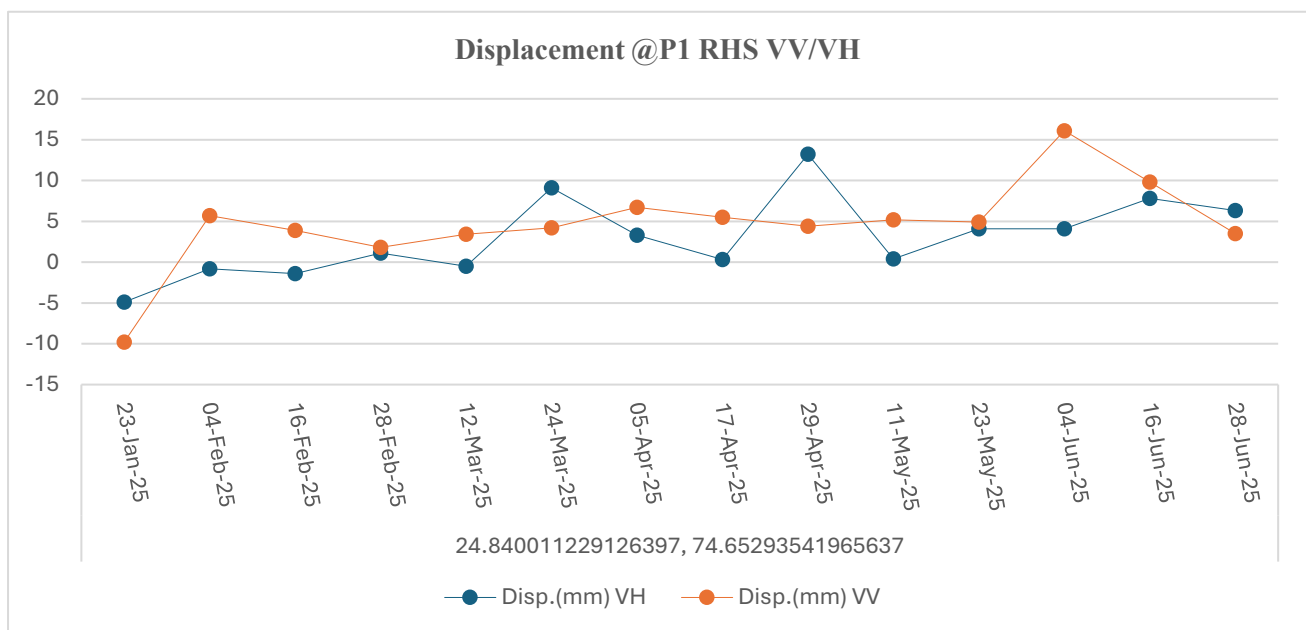
Critical Events & Anomalies:

- **February 2025:** VH transitions from negative to positive, suggesting initial structural realignment or load redistribution.
- **March 2025:** Simultaneous peak VH and maximum downward VV indicate a strong structural response.
- **June 2025:** VH stabilizes at elevated positive values while VV remains consistently negative, suggesting long-term settlement with lateral stabilization.

3.6 SAR Based Analysis at Pier 1 RHS:



P1 RHS (24.840011229126397, 74.65293541965637)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal Displacement (VH)** Horizontal displacement shows moderate variability, generally ranging from approximately –5 mm to +13 mm.
- **Vertical Displacement (VV)** Vertical displacement ranges approximately between –10 mm and +15 mm, showing alternating uplift and settlement phases.

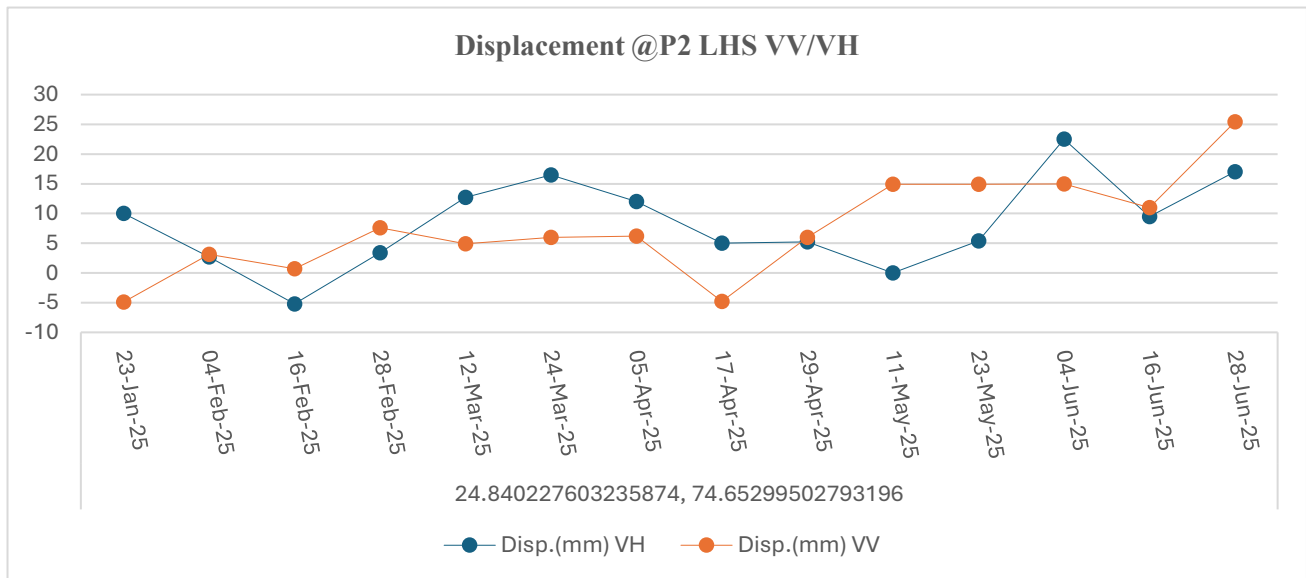
Critical Events & Anomalies:

- **February 2025:** Initial recovery from negative VV values, indicating early-stage structural adjustment.
- **April 2025:** A combined response where VH peaks (+13 mm) while VV remains moderately positive suggesting coordinated horizontal and vertical response rather than distress.
- **June 2025:** Maximum VV uplift (+15 mm), followed by a controlled reduction, indicating temporary vertical expansion rather than permanent deformation.

3.7 SAR Based Analysis at Pier 2 LHS:



P2 LHS (24.840227603235874, 74.65299502793196)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal Displacement (VH)** Horizontal displacement varies between approximately –5 mm and +22 mm, indicating moderate lateral movement throughout the monitoring period.
- **Vertical Displacement (VV)** Vertical displacement ranges approximately between –5 mm and +25 mm, showing both settlement and uplift phases.

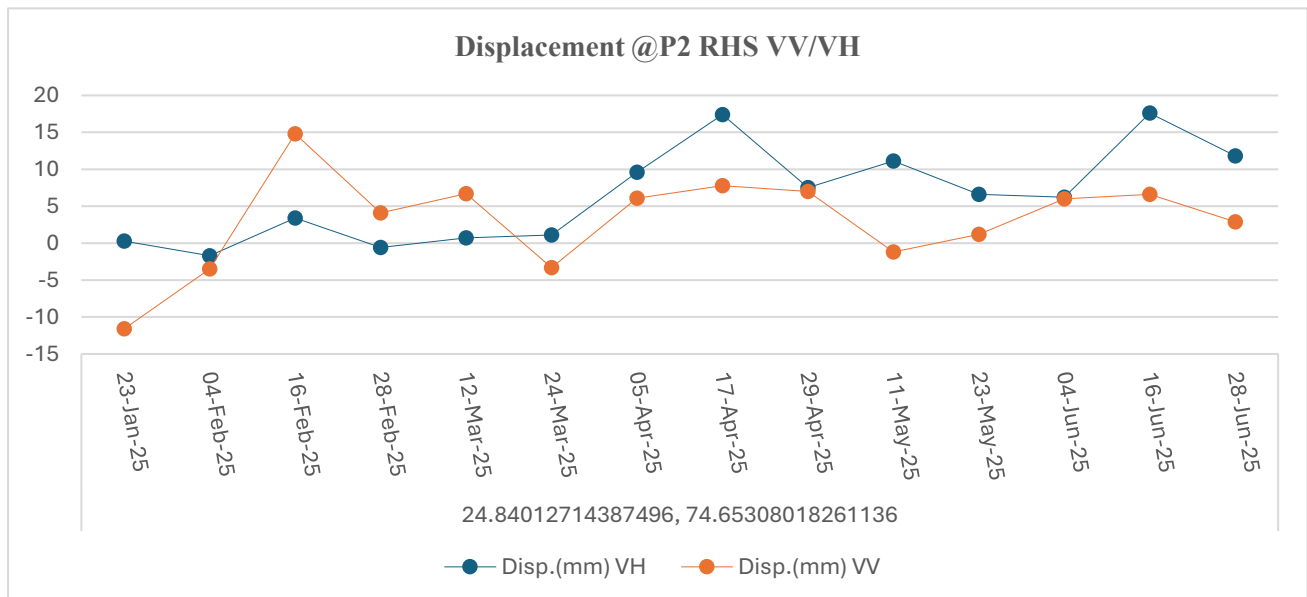
Critical Events & Anomalies:

- **April 2025:** A short-lived vertical drop (–5 mm) accompanied by moderate VH values suggests a temporary structural or environmental influence.
- **June 2025:** Simultaneous peaks in VH (+22 mm) and VV (+25 mm) indicate combined lateral and vertical response, possibly linked to thermal expansion or load redistribution.

3.8 SAR Based Analysis at Pier 2 RHS:



P2 RHS (24.84012714387496, 74.65308018261136)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal Displacement (VH)** Horizontal displacement varies approximately between -2 mm and $+17$ mm, indicating moderate lateral movement throughout the monitoring period.
- **Vertical Displacement (VV)** Vertical displacement ranges approximately from -10 mm to $+15$ mm, exhibiting alternating uplift and settlement phases.

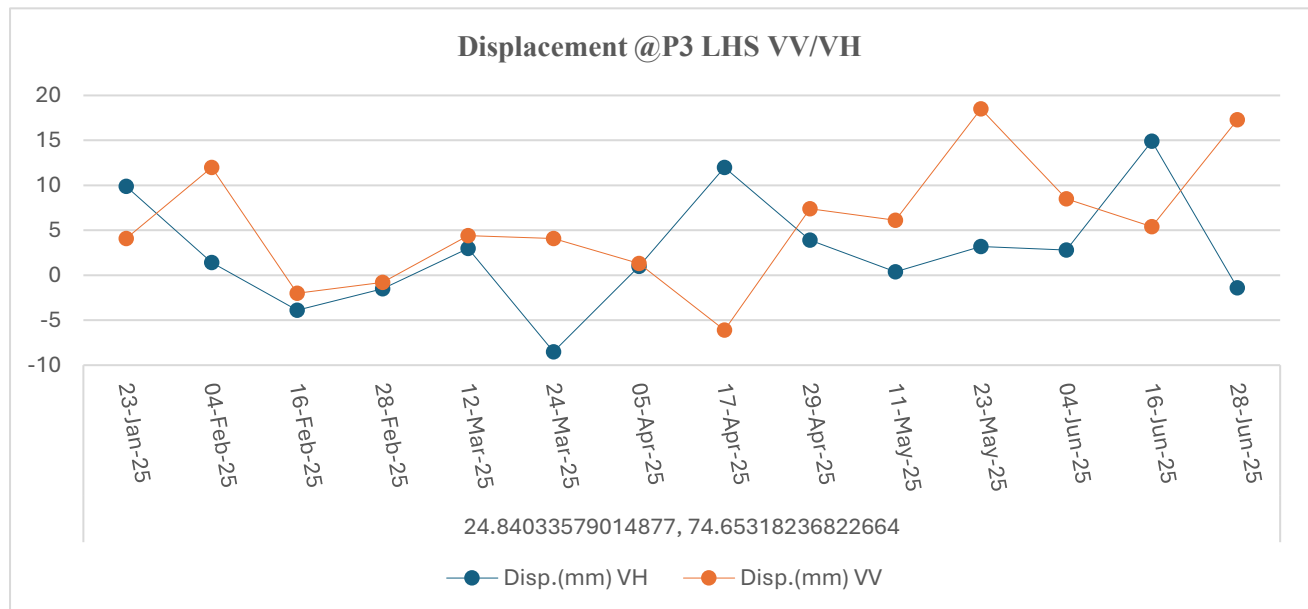
Critical Events & Anomalies:

- **February 2025:** A short-lived VV peak ($+15$ mm) combined with low VH suggests localized vertical response rather than structural instability.
- **April 2025:** VH reaches its highest value ($+17$ mm), indicating a temporary lateral response likely influenced by environmental or operational loading
- **May 2025:** A brief dip in VV (3 mm) followed by steady recovery suggests transient settlement with no progressive deformation

3.9 SAR Based Analysis at Pier 3 LHS:



P3 LHS (24.84033579014877, 74.65318236822664)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal Displacement (VH)** Horizontal displacement varies approximately between –8 mm and +15 mm, showing moderate fluctuation throughout the monitoring period.
- **Vertical Displacement (VV)** Vertical displacement ranges approximately from –6 mm to +18 mm, reflecting alternating settlement and uplift phases.

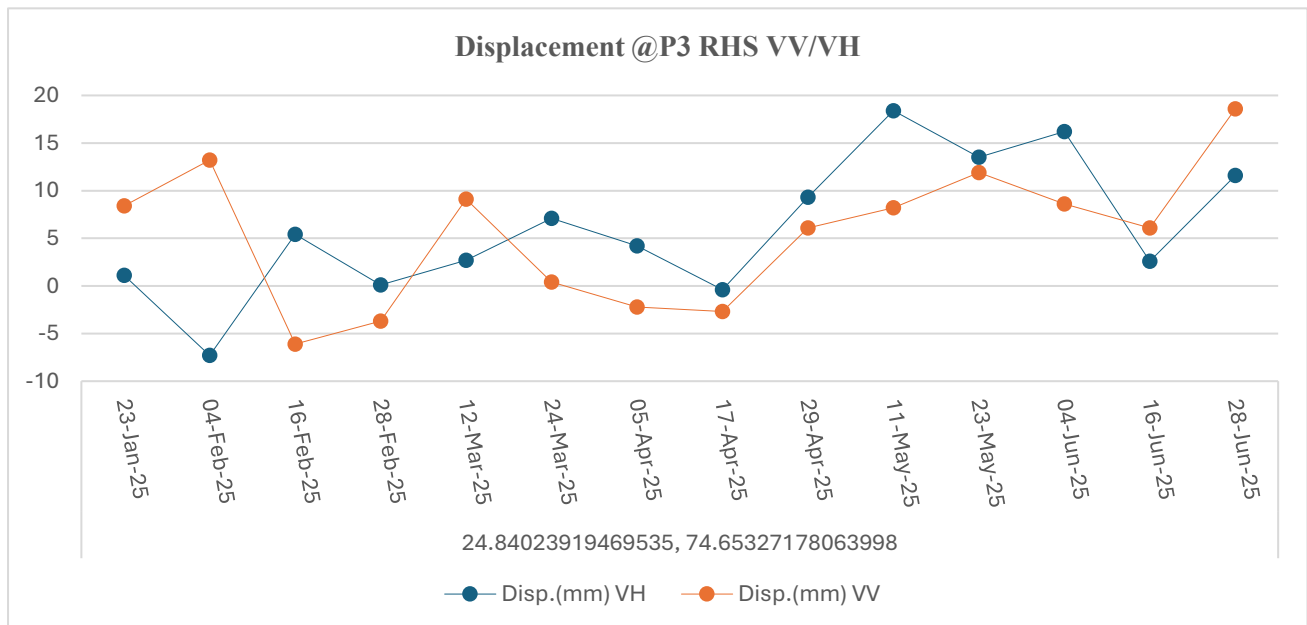
Critical Events & Anomalies:

- **March 2025:** A brief simultaneous reduction in both VH and VV suggests a temporary response, possibly linked to environmental or operational influences.
- **May 2025:** The most significant vertical uplift (+18 mm) occurs, accompanied by moderate horizontal movement indicating a localized but controlled structural response
- **June 2025:** VH peaks again (+15 mm), followed by a short-term reduction, suggesting transient lateral redistribution rather than progressive deformation.

3.10 SAR Based Analysis at Pier 3 RHS:



P3 RHS (24.84023919469535, 74.65327178063998)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal Displacement (VH)** Horizontal displacement ranges approximately between -7 mm and $+18$ mm, showing moderate variability throughout the monitoring period.
- **Vertical Displacement (VV)** Vertical displacement varies between approximately -5 mm and $+18$ mm, reflecting alternating uplift and settlement phases.

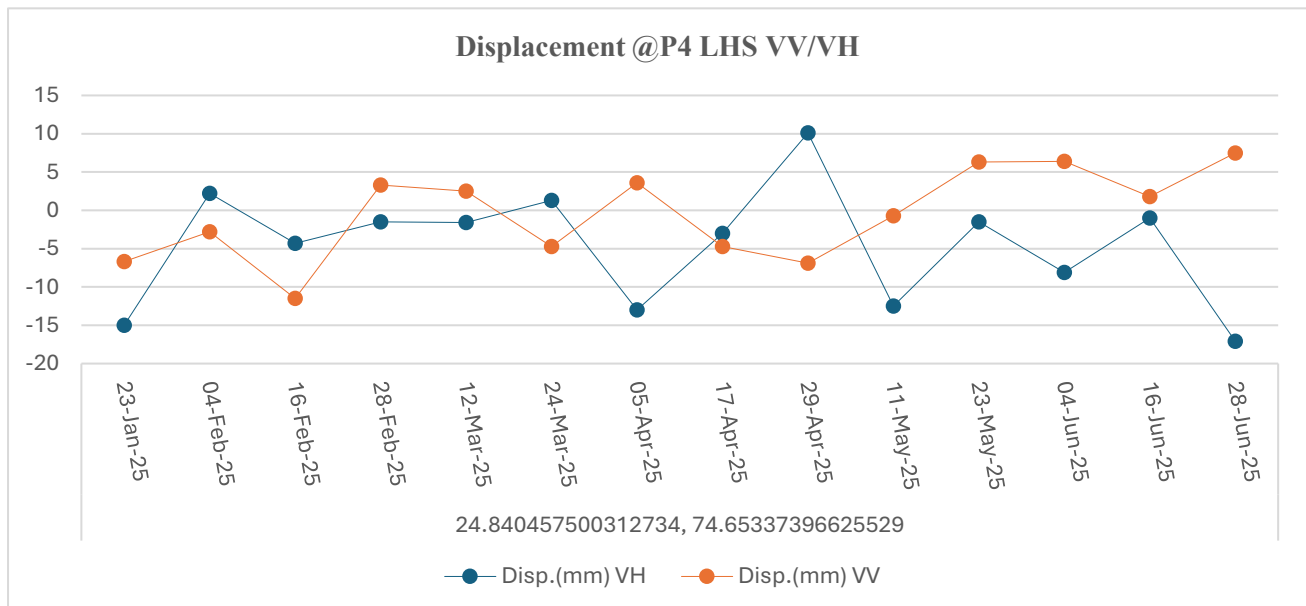
Critical Events & Anomalies:

- **February 2025:** A brief negative VH (-7 mm) coupled with positive VV indicates localized movement, likely transient in nature.
- **April-May 2025:** Both VH and VV rise concurrently, indicating combined lateral and vertical structural response, potentially linked to thermal or load-induced effects
- **June 2025:** Peak VV ($+18$ mm) and elevated VH ($+12$ mm) suggest increased structural activity, though remaining within acceptable displacement limits.

3.11 SAR Based Analysis at Pier 4 LHS:



P4 LHS (24.840457500312734, 74.65337396625529)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal Displacement (VH)** Horizontal displacement generally fluctuates between approximately –18 mm and +10 mm, indicating moderate to high variability over the monitoring period.
- **Vertical Displacement (VV)** Vertical displacement varies roughly between –10 mm and +8 mm, exhibiting alternating settlement and uplift phases. Compared to VH, the VV trend is smoother but still demonstrates notable cyclical behaviour.

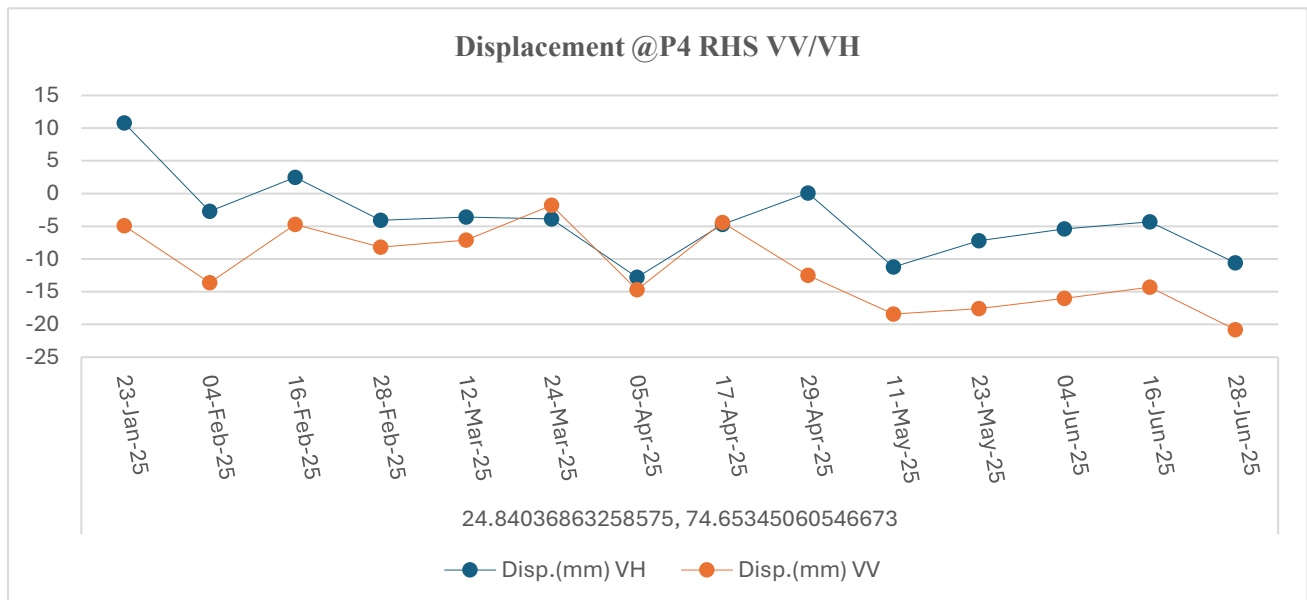
Critical Events & Anomalies:

- **February 2025:** A negative VV excursion (–10 mm) coincides with near-neutral VH, suggesting a short-term settlement event with limited horizontal influence.
- **April 2025:** A distinct VH peak (+10 mm) occurs while VV remains slightly negative, highlighting a dominant horizontal response during this period.
- **June 2025:** VV reaches one of its higher positive values (+7 to +8 mm) while VH trends downward again, ending with a sharp negative VH (–18 mm).

3.12 SAR Based Analysis at Pier 4 RHS:



P4 RHS (24.84036863258575, 74.65345060546673)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal Displacement (VH)** Horizontal displacement ranges approximately between +10 mm and -12 mm, showing moderate variability with an overall declining tendency across the monitoring period.
- **Vertical Displacement (VV)** Vertical displacement varies from about -5 mm to -22 mm, demonstrating a predominantly downward (settlement) trend. Unlike VH, VV shows limited recovery once negative displacement increases, suggesting progressive vertical movement.

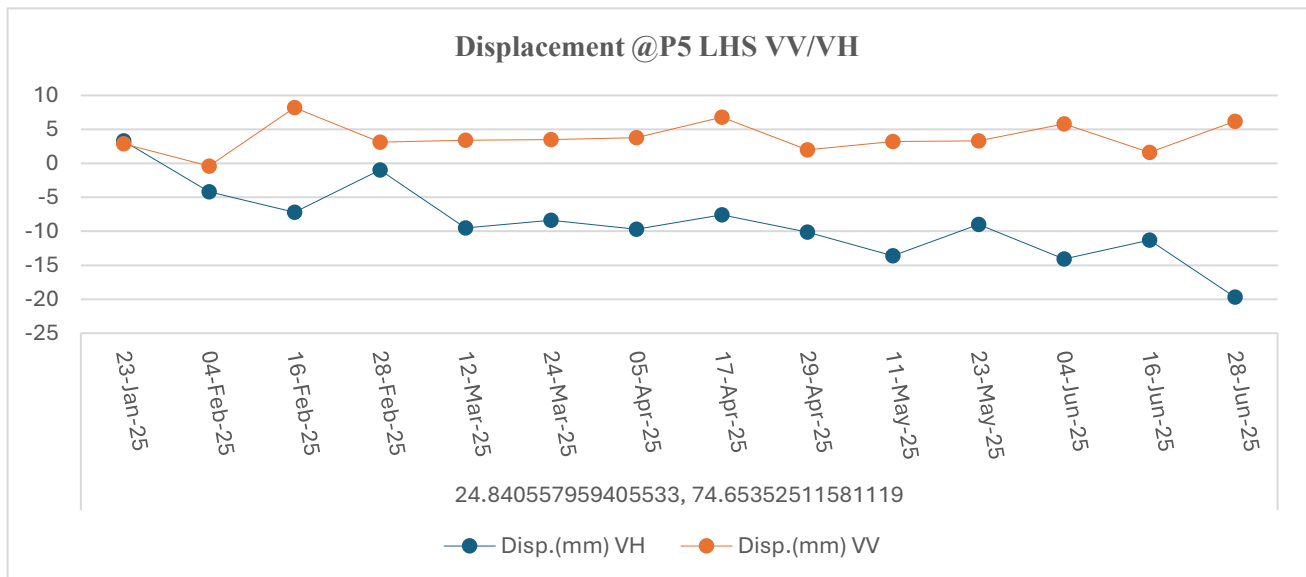
Critical Events & Anomalies:

- **January 2025:** VH records a clear positive peak (+10 mm) while VV is mildly negative (-5 mm), indicating an initial lateral response with minimal vertical settlement.
- **April 2025:** A pronounced simultaneous drop occurs, with VH reaching around -13 mm and VV near -15 mm, indicating a combined lateral and vertical displacement event
- **June 2025:** The monitoring period concludes with sustained negative displacements, with VH around -12 mm and VV reaching its maximum settlement.

3.13 SAR Based Analysis at Pier 5 LHS:



P5 LHS (24.840557959405533, 74.65352511581119)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal Displacement (VH)** Horizontal displacement varies approximately between –20 mm and 0 mm, showing a predominantly negative trend throughout the monitoring period.
- **Vertical Displacement (VV)** Vertical displacement ranges roughly from 0 mm to +8 mm, remaining consistently positive for most of the observation window. This pattern reflects persistent uplift with relatively low variability compared to VH.

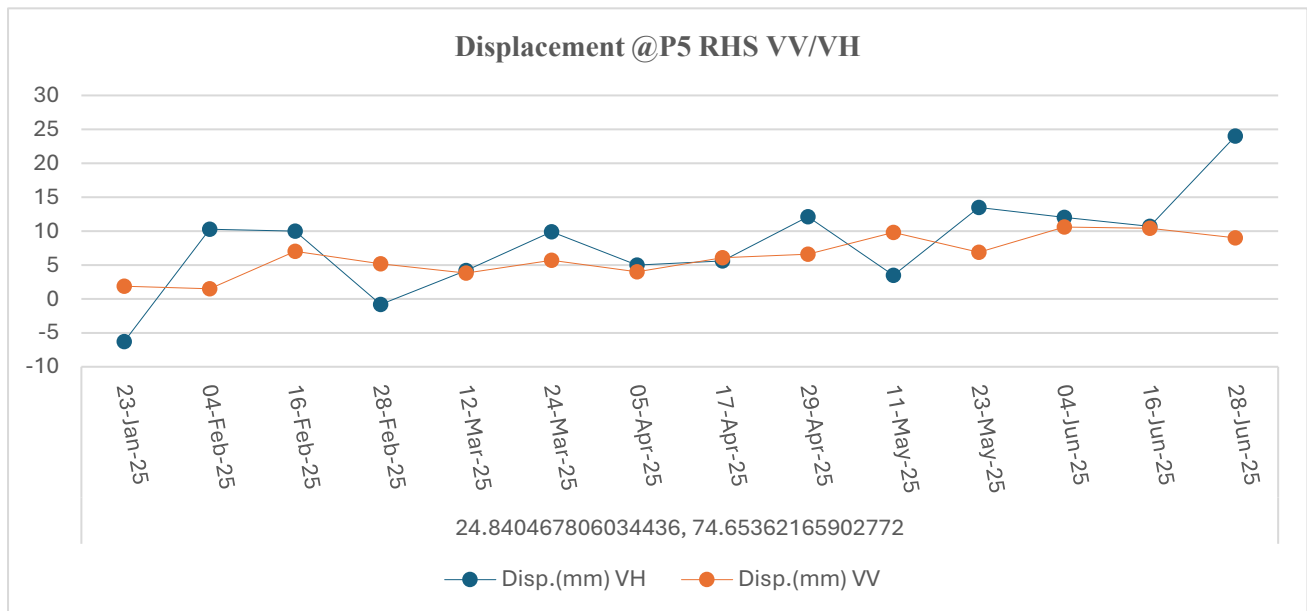
Critical Events & Anomalies:

- **January 2025:** Both VH and VV start near slightly positive to neutral values, suggesting an initially stable condition with minimal displacement.
- **May 2025:** VH continues its downward progression, reaching approximately –13 mm, whereas VV remains moderately positive indicating continued lateral displacement with stable vertical uplift
- **June 2025:** The period ends with the maximum negative VH and VV recovering to around +6 to +7 mm, reflecting intensified lateral movement while vertical displacement remains controlled and non-progressive.

3.14 SAR Based Analysis at Pier 5 RHS:



P5 RHS (24.840467806034436, 74.65362165902772)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal Displacement (VH)** displacement ranges approximately from -7 mm to +24 mm over the monitoring period.
- **Vertical Displacement (VV)** VV displacement varies roughly between +2 mm and +11 mm. The values remain consistently positive throughout, indicating sustained vertical uplift.

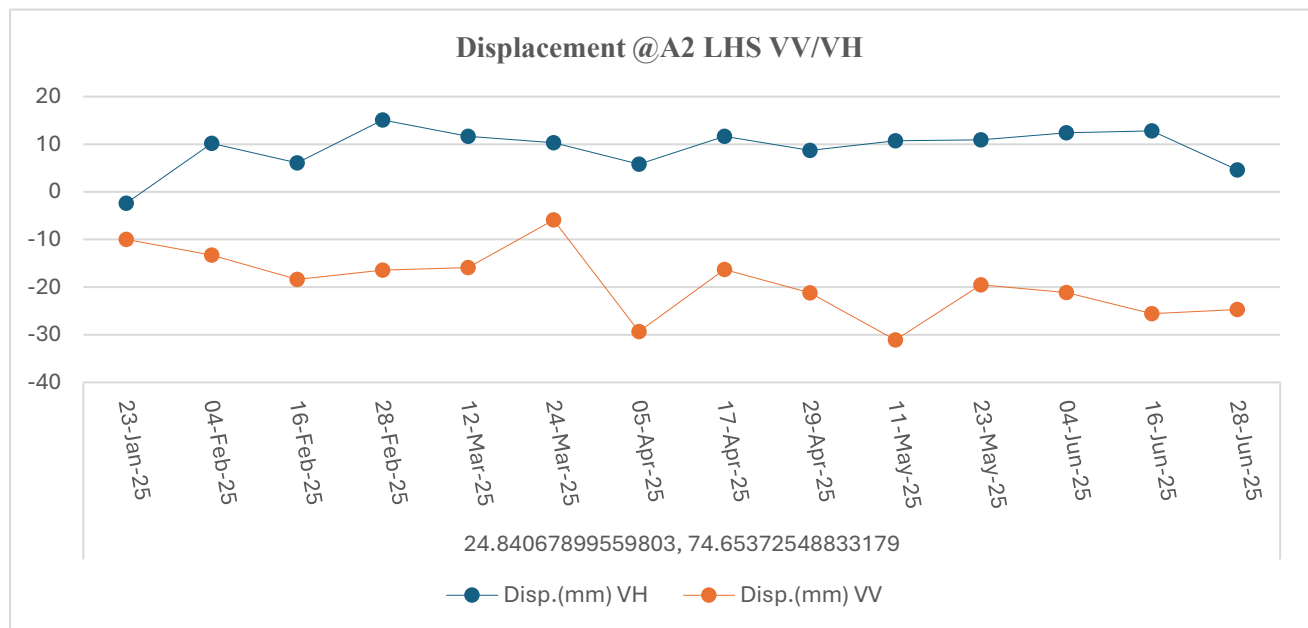
Critical Events & Anomalies:

- **January 2025:** VH begins slightly negative before quickly shifting to positive values near +10 mm. VV remains low but positive (2–3 mm), indicating an initially stable vertical condition
- **May 2025:** VH displays renewed lateral acceleration, reaching approximately +13–14 mm. VV increases further to about 8–10 mm, indicating continued but controlled vertical uplift.
- **June 2025:** VH reaches its maximum recorded value, marking the most significant horizontal displacement event in the dataset. VV remains comparatively stable around 9–10 mm,

3.15 SAR Based Analysis at Abutment 2 LHS:



A2 LHS (24.84067899559803, 74.65372548833179)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal Displacement (VH)** VH displacement ranges approximately from +3 mm to +15 mm. VH shows moderate variability, with an overall stable to slightly increasing trend toward June 2025
- **Vertical Displacement (VV)** VV displacement ranges roughly from –10 mm to –32 mm. The displacement remains consistently negative, indicating persistent vertical settlement

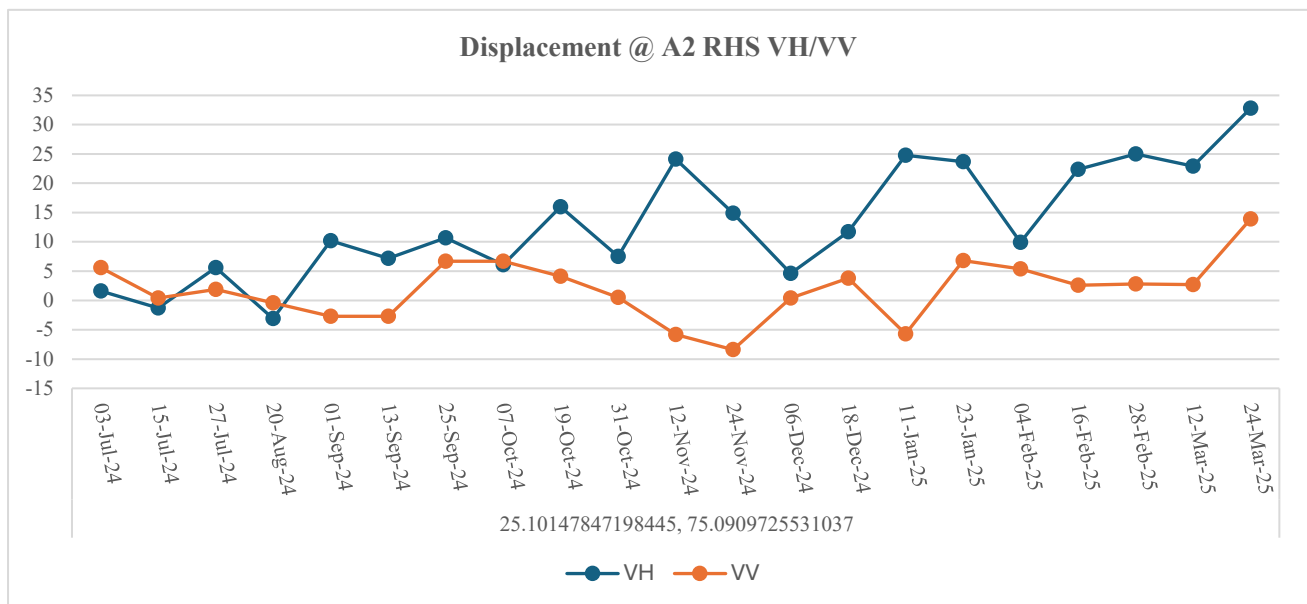
Critical Events & Anomalies:

- **January 2025:** Both VH and VV start with moderate displacement values. VH is slightly positive at approximately +3 to +5 mm, indicating limited lateral movement, while VV begins around –10 mm
- **May 2025:** VH maintains a consistent value near +10 to +11 mm, indicating stable lateral behaviour.
- **June 2025:** The monitoring period concludes with VH fluctuating between +5 mm and +13 mm, showing minor lateral adjustment without instability.

3.16 SAR Based Analysis at Abutment 2 LHS:



A2 RHS (24.840601017954985, 74.65382215699417)



Key Observations:

Displacement Magnitude Patterns:

- **Horizontal Displacement (VH)** Horizontal displacement exhibits high variability with alternating positive and negative phases, ranging roughly between +8 mm and -12 mm.
- **Vertical Displacement (VV)** Vertical displacement remains predominantly positive throughout the monitoring period (Jan–Jun 2025), indicating overall vertical uplift at the A2 RHS location

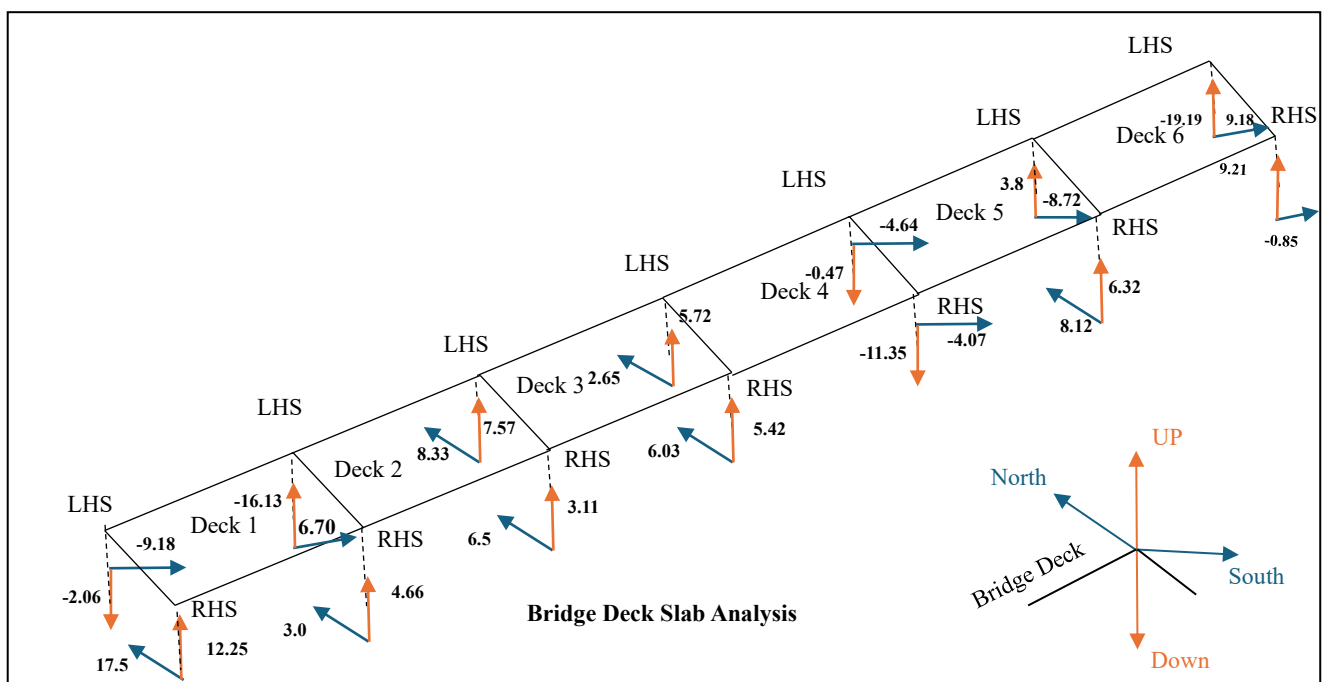
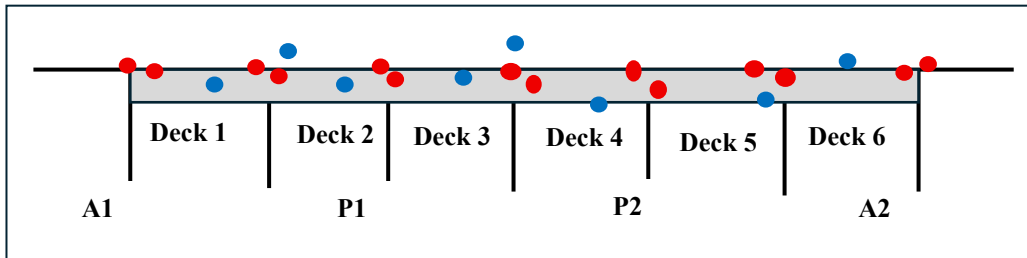
Critical Events & Anomalies:

- **February 2025:** VH reaches a local positive peak (+8 mm), while VV drops to +2 mm, marking a brief coupled stabilization phase
- **April 2025:** VH declines again to -8 mm, while VV rises, marking another decoupled deformation phase
- **June 2025:** VH remains mostly negative (-3 to -10 mm), while VV stays elevated (+10–15 mm), confirming continued uplift alongside unresolved lateral instability toward the end of the monitoring period.

ANALYSIS

1. Identify potential deformation, shifts or movement of the bridge structure

- Points of Analysis (POA) of bridge (●)



Vertical Displacement Behaviour:

- The bridge deck system exhibits strongly non-uniform vertical displacement along its longitudinal length, indicating differential vertical response across Decks 1 to 6. The maximum downward settlement is observed at Deck 6 (RHS), reaching approximately -19.19 mm, representing the most critical vertical deformation in the entire deck sequence

Horizontal Displacement Behaviour:

- Horizontal displacement varies significantly along the deck length and across LHS-RHS, indicating complex torsional and translational behaviour. The maximum horizontal displacement is observed near Deck 6 (RHS) at approximately +9.21 mm, suggesting dominant lateral movement at the terminal deck region. The presence of alternating horizontal displacement directions strongly indicates distributed torsional rotation rather than rigid-body translation

CONCLUSION:

- The bridge deck slab system demonstrates a highly complex deformation pattern, characterized by alternating vertical uplift and settlement across successive decks. The magnitude and variability of displacements, particularly near Deck 6, suggest structurally significant behaviour.